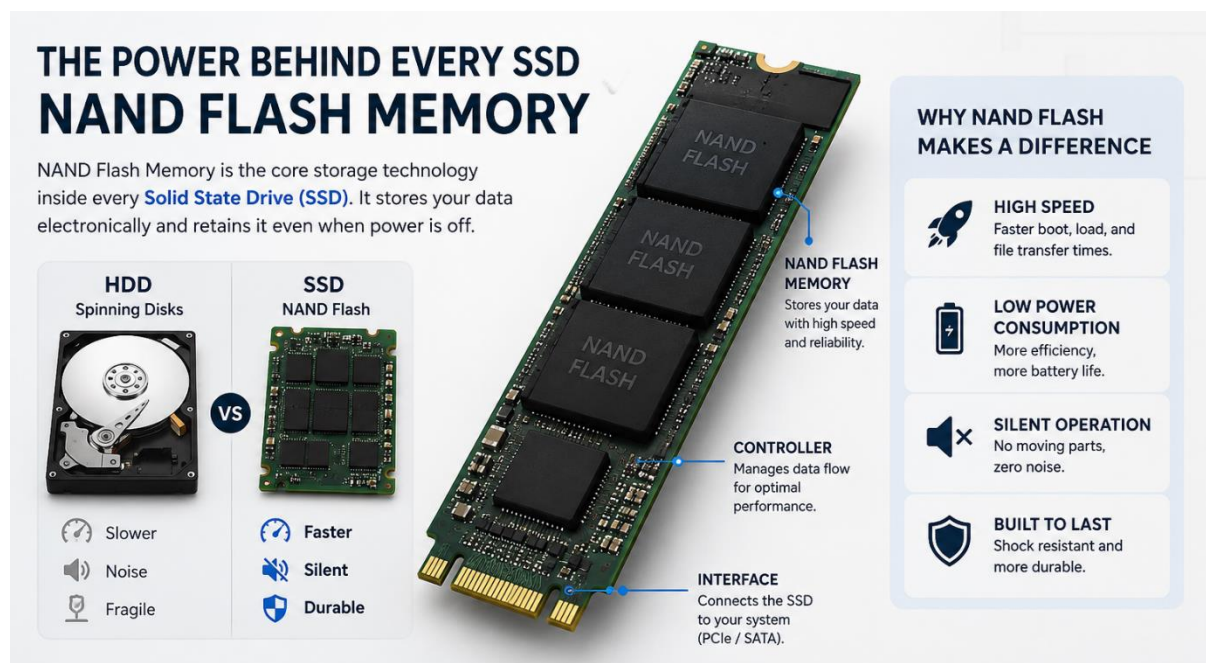


Understanding NAND Flash Memory: The Heart of Every SSD

What is NAND Flash Memory?

NAND Flash Memory is the core storage technology used in every Solid State Drive (SSD). As a non-volatile memory, it safely stores your data even when the power is off. Unlike traditional HDDs with spinning disks, NAND flash delivers faster speeds, lower power consumption, silent operation, and greater durability. At **PARTPIXELS**, we believe understanding SSD technology helps you choose the right storage solution with confidence.



Types of NAND Flash Memory

SLC (Single-Level Cell)

Stores: 1 bit per cell

SLC is the fastest and most reliable NAND type, offering the highest endurance and longest lifespan. It is expensive, so it is mainly used in enterprise, industrial, and mission-critical applications.

MLC (Multi-Level Cell)

Stores: 2 bits per cell

MLC provides a good balance of speed, durability, and storage capacity. It is commonly used in professional workstations and business systems that require reliable performance.

TLC (Triple-Level Cell)

Stores: 3 bits per cell

TLC is the most popular NAND type in modern consumer SSDs. It offers an excellent balance of performance, endurance, and affordability, making it ideal for gaming, office work, and everyday use.

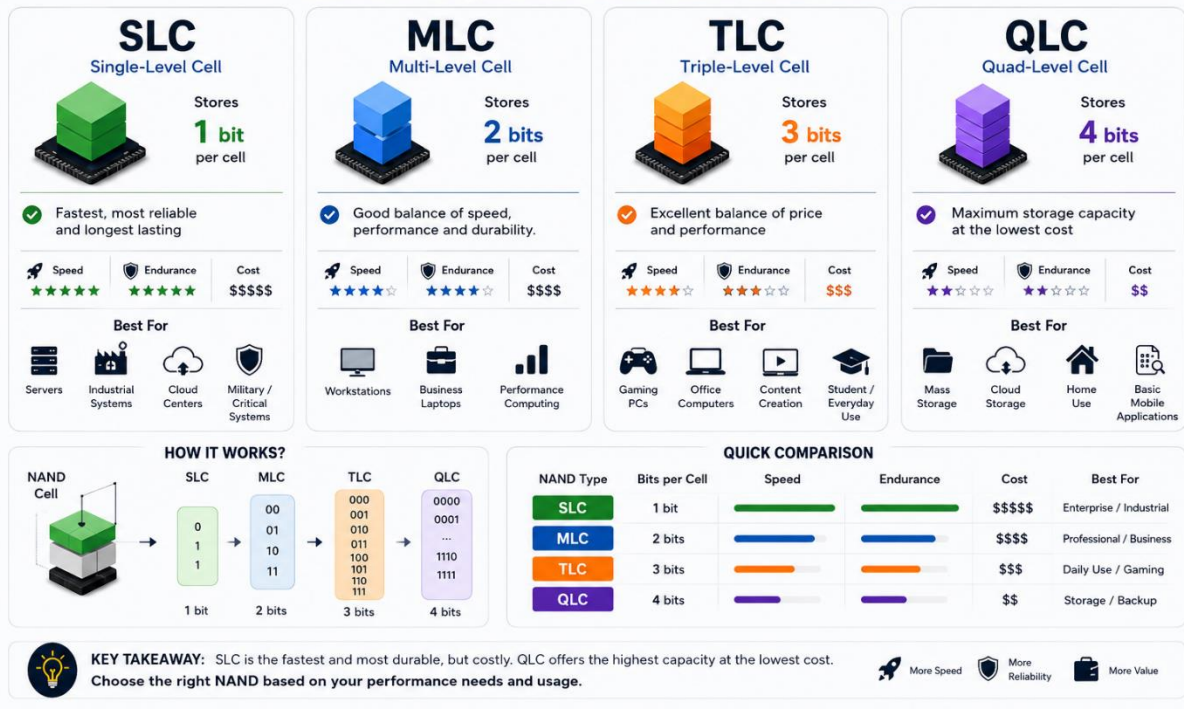
QLC (Quad-Level Cell)

Stores: 4 bits per cell

QLC delivers the highest storage capacity at the lowest cost per GB. It is best suited for storing large files, backups, and media libraries, but is less suitable for heavy write-intensive workloads.

UNDERSTANDING NAND FLASH MEMORY TYPES

Different NAND types store different amounts of data in each cell.



Common Myths About NAND

Myth: QLC SSDs are unreliable.

Reality: QLC SSDs are reliable for many every day and read-heavy applications, though they generally have lower endurance than TLC SSDs.

Myth: Higher advertised speed means a better SSD.

Reality: Overall performance depends on controller design, firmware, sustained performance, and workload—not just peak sequential speeds.

Myth: All TLC SSDs perform the same.

Reality: Different controllers, firmware, cache designs, and NAND quality can lead to significant performance differences between TLC SSDs.

Which NAND type is best?

There is no single "best" NAND type. The right choice depends on your workload. TLC is the preferred option for most users because it offers an excellent balance of speed, endurance, and cost.

Is TLC better than QLC?

For users who frequently write large amounts of data, TLC generally offers better sustained performance and endurance. QLC excels when maximizing storage capacity at a lower cost is the priority.

How long does a TLC SSD last?

For typical home and office use, a quality TLC SSD can provide many years of service before reaching its rated endurance.

Can NAND wear out?

Yes. Every NAND cell has a finite number of program/erase cycles. Modern SSDs use wear leveling, over-provisioning, error correction, and spare blocks to maximize lifespan and maintain reliability.